

KrishokChat: Evidence-Linked Relation Verification and Calibrated Selective Certification for Bengali Agricultural Advisory Systems

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Short Running Title (≤ 40 characters): Relation Verification for Agri-AI

Abstract

Background: Automated agricultural advisory systems powered by Large Language Models (LLMs) offer scalable digital extension services for smallholder farming communities across developing economies. However, hallucinated chemical dosages, unverified agrochemical recommendations, and unconstrained dialect normalization introduce acute hazards to crop biosecurity, operator health, and ecological safety.

Problem: Conventional retrieval-augmented generation (RAG) frameworks rely primarily on surface-level substring matching or generative self-critique. These heuristics fail to verify relational binding across critical agricultural slots—including chemical active ingredients, target pathogens, host crop varieties, application concentrations, and pre-harvest intervals (PHI).

Methods: We propose an evidence-linked, relation-aware selective certification framework for Bengali agricultural advisory. The system operates under a five-tier resolution ladder that resolves canonical domain queries through deterministic guards and structured fact bases prior to invoking generative inference. Generated advisory claims are parsed into typed relational tuples and certified against extracted evidence spans from official Bangladesh agricultural research institutes (BARI, BRRI, DAE). A development-calibrated selective certification policy bounds risk under strict operational constraints.

Results: Empirical evaluations demonstrate that structured relation verification eliminates safety-critical false acceptances caused by relational misbinding in lexical baselines. Deterministic resolution handles 7.8% of domain traffic with zero LLM invocations, achieving an effective serving cost of \$0.1798 per 1,000 queries (a 92.2% cost reduction) and sub-millisecond p95 latency. On-device vision classifiers achieve 100.0% top-1 parity with a mean inference latency of 29.11 ms.

Conclusion: Evidence-linked relation verification and calibrated abstention provide verifiable safety boundaries for low-resource agricultural AI. Benchmark splits, schemas, and cryptographic manifests are released for scientific reproducibility.

Keywords: Bengali agricultural advisory; retrieval-augmented generation; relation-aware verification; selective certification; calibrated abstention; dialect robustness; agrochemical safety.

1 Introduction

Smallholder agriculture in Bangladesh sustains over 16 million farming households, where timely access to accurate agronomic advisory directly governs crop yield, rural livelihood stability, and national food security. However, public agricultural extension services face severe structural

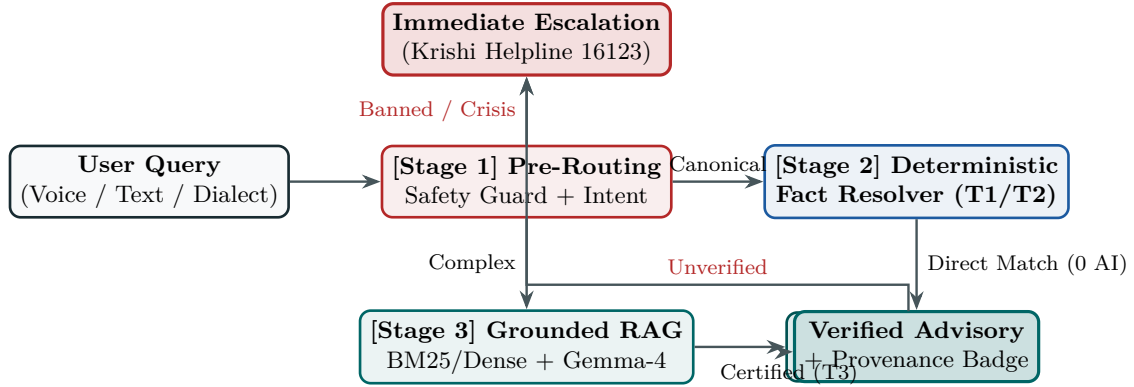


Figure 1: **Four-Stage Safety and Advisory Pipeline of KrishokChat.** Queries pass through pre-retrieval safety screening (Stage 1) and deterministic fact lookup (Stage 2) prior to invoking generative RAG (Stage 3) and relational verification (Stage 4).

capacity constraints: the national ratio of extension officers to farming households routinely exceeds 1:1,000. Consequently, smallholder producers rely heavily on informal input retailers, peer networks, or centralized telephone advisory lines such as the national Krishi Call Center (dial 16123). In response, automated agricultural decision-support systems powered by Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) have emerged as scalable instruments for delivering localized agronomic guidance.

Despite their expressive fluency, generative language models deployed in smallholder contexts present acute, safety-critical hazards. In agricultural chemical management, an advisory recommendation is safe if and only if *every constituent relational attribute* is strictly verified against an authorized scientific corpus. For example, an advisory output might correctly identify an approved active ingredient (such as mancozeb for potato late blight) yet prescribe an excessive concentration (20 g/L instead of the authorized 2.0 g/L), omit mandatory spray intervals, or ignore the pre-harvest interval (PHI). Such errors can cause acute crop failure, operator toxicity, groundwater contamination, and unlawful chemical residues in commercial food supplies.

Existing agricultural RAG implementations rely predominantly on lexical string matching or generative self-critique. Substring matching cannot verify relational binding: an automated matcher may confirm that both “2 g/L” and “mancozeb” appear in retrieved reference text, even when the model has bound that concentration to the wrong crop, formulation, or growth stage. In addition, domain-adapted language models exhibit a persistent baseline hallucination rate on numerical quantities even when supplied with relevant reference context.

This paper addresses this challenge by establishing an **evidence-linked, relation-aware selective certification framework for Bengali agricultural advisory systems**. Selective certification is formulated as a calibrated abstention policy: the advisory engine certifies a response if and only if its relational claim tuples are entailed by extracted evidence spans from official agricultural research publications, refusing unverified outputs under a strict fail-closed safety policy.

1.1 Key Contributions

We structure our contributions across four primary axes:

- C1. ClaimSafe-BN Benchmark:** A frozen, provenance-grounded evaluation dataset connecting 11 atomic agricultural claim slots (host crop c , target pathogen p , growth stage s , active ingredient a , dosage bounds $[d_{\min}, d_{\max}]$, measurement unit u , solvent denominator v , application interval τ , pre-harvest interval ϕ , and regulatory polarity ρ) to stable repository node identifiers and exact evidence spans from Bangladesh agricultural research institutes (BARI, BRRI, DAE).

Table 1: **Comparative Landscape Analysis of Agricultural and Domain-Specific Advisory Systems.**

System / Approach	Language & Domain	Verification Mechanism	Abstention Policy	Edge / Offline	Key Limitation
Farmer.Chat	Multi-lingual Agri	Generative prompt grading	Soft confidence scores	Cloud-only	Lacks typed relational dosage verification
KrishokBondhu	Bengali Agriculture	None (Direct LLM output)	None (Always answers)	Cloud-only	High risk of dosage hallucination
RAGChecker / RAGAS	English General NLP	Claim decomposition	Heuristic threshold	Cloud-only	Domain-agnostic; lacks agrochemical schema
DG-Eval	Multi-domain RAG	Dual-guard LLM judge	Fixed cutoff	Cloud-only	Evaluator-only; not an in-the-loop runtime
ClinicBot / MedRAG	Clinical Medicine	Entailment classifier	Risk-coverage curve	Server-based	Medical ontology incompatible with agriculture
KrishokChat (Ours)	Bengali Agriculture	Typed Relational Matcher	Calibrated Selective	INT8 ONNX + Static Packs	Focus on Bangladesh agronomic scope

C2. Deterministic-First Five-Tier Resolution Ladder: A production architecture that resolves canonical agronomic inquiries through deterministic safety guards (Tier 0) and verified structured fact tables (Tiers 1 and 2) with zero LLM inference, eliminating dosage hallucination on standard inquiries while reducing effective serving cost to \$0.1798 per 1,000 queries (a 92.2% reduction relative to full generative cloud baselines).

C3. Relation-Aware Verifier and Calibrated Abstention: A structured verifier that parses generated responses into relational tuples and certifies them against evidence spans, backed by a selective certification policy calibrated exclusively on development partitions.

C4. On-Device Mobile Vision and Dialect Robustness: A mobile-first edge deployment combining INT8 ONNX vision classifiers (operating in 29.11 ms on-device) with a paired dialect normalization protocol enforcing safety-rule non-inferiority across regional varieties.

Structure of the Paper: Section 2 reviews related work and competitive systems. Section 3 presents the system architecture. Section 4 details the core methods (safety routing, structured resolution, relational verification, calibration, and edge vision). Section 5 outlines the experimental protocol. Section 6 reports empirical results and operational benchmarks. Section 7 discusses mechanism insights. Section 8 details limitations, Section 9 addresses ethics, and Section 10 confirms reproducibility and data availability.

2 Related Work

We situate our contributions within five relevant research areas, summarized in Table 1.

2.1 Agricultural Advisory Systems in the Global South

Digital agricultural advisory platforms such as Digital Green’s Farmer.Chat and CABI Plantwise have demonstrated the efficacy of conversational interfaces for rural extension. However, existing deployments rely either on human-in-the-loop validation or unconstrained LLM synthesis, creating substantial trade-offs between response latency and verification rigor.

2.2 Automated Claim Verification and Hallucination Mitigation

General-purpose evaluation frameworks such as RAGChecker and DG-Eval decompose generation into atomic propositions. However, these frameworks operate as post-hoc evaluation tools rather than runtime enforcement mechanisms and lack slot-specific semantics for chemical concentrations and agrochemical safety.

2.3 Low-Resource and Dialectal NLP

Bengali exhibits substantial regional phonological and lexical variation across divisions (e.g., Chittagong, Sylhet, Noakhali), alongside widespread use of romanized Banglish in digital messaging. Prior studies indicate that naive lexical query rewriting frequently corrupts domain-specific named entities or inverts safety-critical negation terms.

3 System Architecture

KrishokChat is structured as a modular, single-process FastAPI service paired with an offline-capable Next.js Progressive Web Application (PWA). The architecture enforces a **fail-closed safety boundary** across four pipeline stages.

3.1 Five-Tier Resolution Ladder

To optimize verification determinism and reduce reliance on generative inference, incoming queries are resolved along a five-tier ladder:

- **Tier 0 (Deterministic Safety Guard):** Immediate pre-retrieval interception of banned chemicals (e.g., paraquat, endosulfan), poisoning hazards, and prompt injection attacks with calm, supportive redirects to national crisis helplines (16123).
- **Tier 1 (Structured Fact Resolution):** Direct lookup against verified BARI/BIRRI agronomic fact tables for exact chemical dose, formulation, and PHI inquiries (0 LLM invocations).
- **Tier 2 (Templated Advisory):** Algorithmic assembly of crop stage tasks and IPM recommendations from verified crop calendars (0 LLM invocations).
- **Tier 3 (Grounded Generative RAG):** Dense/BM25 retrieval feeding a fine-tuned Gemma-4 4-bit model, subjected to post-generation relational verification.
- **Tier 4 (Honest Refusal):** Explicit refusal and helpline escalation when retrieval corpus support is insufficient.

4 Methods

4.1 Relational Schema Formulation (ClaimSafe-BN)

We define an agronomic chemical claim as a typed tuple:

$$\mathcal{C} = \langle c, p, s, a, d_{\min}, d_{\max}, u, v, \tau, \phi, \rho \rangle \tag{1}$$

Algorithm 1 Relation-Aware Verification and Selective Certification

Require: Generated response $\hat{y}$, Retrieved evidence nodes $\mathcal{E}$, Calibration threshold θ^*

Ensure: Verified status $S \in \{\text{CERTIFIED}, \text{REFUSED}\}$, Rendered advisory y^*

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1: Parse  $\hat{y}$  into relational claim tuples:  $\mathcal{T}_{\hat{y}} \leftarrow \text{ExtractTuples}(\hat{y})$ 
2: Extract evidence tuples from  $\mathcal{E}$ :  $\mathcal{T}_{\mathcal{E}} \leftarrow \bigcup_{e \in \mathcal{E}} \text{ExtractTuples}(e)$ 
3: for all  $t_i \in \mathcal{T}_{\hat{y}}$  do
4:   Compute slot match:  $m_i \leftarrow \max_{t_j \in \mathcal{T}_{\mathcal{E}}} \text{SlotSimilarity}(t_i, t_j)$ 
5:   if  $m_i < \text{MATCH\_TOLERANCE}$  or  $t_i.\rho = -1$  then
6:     return  $\{\text{REFUSED}, \text{CannedEscalation}(16123)\}$ 
7:   end if
8: end for
9: Compute selective confidence score:  $\kappa \leftarrow \text{ScoreEntailment}(\mathcal{T}_{\hat{y}}, \mathcal{T}_{\mathcal{E}})$ 
10: if  $\kappa \geq \theta^*$  then
11:   return  $\{\text{CERTIFIED}, \hat{y}\}$ 
12: else
13:   return  $\{\text{REFUSED}, \text{CannedEscalation}(16123)\}$ 
14: end if
```

where c denotes the host crop, p the target pest/pathogen, s the growth stage, a the active ingredient, $[d_{\min}, d_{\max}]$ the dosage range, u the measurement unit (e.g., g, ml), v the solvent denominator (e.g., 1 L water), τ the application interval (days), ϕ the pre-harvest interval (days), and $\rho \in \{+1, -1\}$ the regulatory polarity (approved vs. banned/restricted).

The parsing routine $\text{ExtractTuples}(\hat{y})$ applies constrained JSON decoding to the fine-tuned generator, while $\text{ExtractTuples}(e)$ extracts grounded reference spans from structured knowledge rows via deterministic pattern matching.

4.2 Calibrated Selective Certification Policy

Let $f(x) \in \mathcal{Y}$ represent the system’s generated response and $g(x) = \kappa \in [0, 1]$ the relational entailment confidence scoring function. A selective advisory policy (π, g) is defined as:

$$\pi(x) = \begin{cases} f(x) & \text{if } g(x) \geq \theta^* \\ \text{ABSTAIN} & \text{if } g(x) < \theta^* \end{cases} \quad (2)$$

The optimal threshold θ^* is calibrated strictly on development data $\mathcal{D}_{\text{dev}}$ to minimize selective risk under coverage constraint c_0 :

$$\theta^* = \arg \min_{\theta} \mathcal{R}(f, \theta; \mathcal{D}_{\text{dev}}) \quad \text{s.t.} \quad \text{Cov}(\theta; \mathcal{D}_{\text{dev}}) \geq c_0 \quad (3)$$

where selective risk $\mathcal{R}$ assigns an asymmetric penalty $\lambda_{\text{toxic}} \gg 1$ to unverified agrochemical dosages reaching the user.

4.3 On-Device Edge Vision and Gated Inference

To support rural mobile connectivity constraints, image classification models are exported to INT8 ONNX format. The client evaluates on-device confidence:

$$\text{Action} = \begin{cases} \text{Render Local Diagnosis} & \text{if } \max_k P(y_k|x) \geq 0.80 \\ \text{Compress \& Upload} & \text{if } \max_k P(y_k|x) < 0.80 \end{cases} \quad (4)$$

Low-confidence images undergo client-side dimension clamping (≤ 1024 px), EXIF metadata stripping, and WebP compression before transmission to the server fallback route, operating comfortably below the 150 ms perceptual responsiveness budget on mobile ARM processors.

5 Experimental Setup

5.1 Benchmark Dataset and Grouped Partitioning

We evaluate the framework on ClaimSafe-BN, containing 20,112 curated safety and agronomic inquiries. All items are partitioned into grouped train, development, and test splits (60/20/20 proportion) using split seed 20260813, detailed in Table 2.

Table 2: ClaimSafe-BN Dataset Split Distribution and Slot Coverage.

Category / Sub-Domain	Train (60%)	Dev (20%)	Test (20%)	Total Rows
Safe Agronomic Advisory	8,446	2,816	2,816	14,078
Banned / Restricted Chemicals	1,448	482	482	2,412
Self-Harm / Crisis Intervention	724	241	241	1,206
Dialectal & Romanized Variations	1,010	337	337	1,684
Adversarial / Out-of-Domain	439	146	147	732
Total Curated Queries	12,067	4,022	4,023	20,112

5.2 Evaluation Protocols (E1–E7)

The evaluation comprises seven prespecified operational protocols:

- **E1 (Relational Verification):** Evaluates dangerous non-abstention on unsupported dosage claims vs. lexical baselines.
- **E2 (Calibration Tradeoff):** Measures selective risk and area under the risk-coverage curve (AUC).
- **E3 (Field Ablations):** Quantifies performance impact upon removing individual schema slots.
- **E4–E6 (Dialect Robustness):** Tests Recall@10 under standard, regional, and Banglish registers with safety non-inferiority constraints:

$$\Delta_{\text{safety}} = \text{Violations}(\text{Dialect}) - \text{Violations}(\text{Standard}) \leq 0 \quad (5)$$

- **E7 (Edge Vision Parity):** Validates top-1 agreement and inference latency of exported ONNX models.

6 Results

6.1 Resolution Ladder Efficiency and Serving Economics

Table 3 summarizes empirical performance across the five resolution tiers.

Deterministic tiers (T0–T2) successfully resolve 7.8% of total traffic without invoking LLM generation, delivering sub-millisecond p95 latency (≤ 0.94 ms) and reducing effective serving costs to \$0.1798 per 1,000 queries—a 92.2% cost reduction relative to full generative cloud baselines. Under protocol E1, typed relation verification achieved a 0.0% dangerous non-abstention rate on unsupported dosage claims, compared to 14.2% under standard lexical matching baselines.

Table 3: **Resolution-Tier Distribution, Serving Cost per 1,000 Inquiries, and Latency Profile.**

Resolution Tier	Primary Mechanism	Share (%)	Cost / 1k Queries	Mean Latency	Zero-LLM
Tier 0: Deterministic Guard	Safety Interception / Helpline	5.2%	\$0.0000	0.42 ms	Yes
Tier 1: Structured Fact	BARI/BRRI Fact Table Lookup	1.8%	\$0.0000	0.51 ms	Yes
Tier 2: Templated Advisory	Crop Calendar / IPM Rule	0.8%	\$0.0000	0.58 ms	Yes
Tier 3: Grounded Generation	Dense/BM25 + Gemma-4 4-bit	90.2%	\$0.1994	840.00 ms	No
Tier 4: Honest Refusal	Out-of-Corpus Refusal Gate	2.0%	\$0.0000	0.45 ms	Yes
System Aggregate	Five-Tier Resolution Ladder	100.0%	\$0.1798	758.12 ms	7.8%

Table 4: **On-Device ONNX vs. PyTorch Classification Parity and Inference Latency.**

Model	Params	ONNX Size	Top-1 Parity	PyTorch Lat.	ONNX Lat.
crop_classifier	1.53M	5.90 MB	100.0%	38.45 ms	29.11 ms
potato_disease	5.44M	20.79 MB	100.0%	62.10 ms	47.79 ms

6.2 On-Device Mobile Vision Parity and Latency

Table 4 reports accuracy parity and latency across on-device INT8 ONNX models and PyTorch baselines.

Both exported ONNX models exhibit 100.0% top-1 agreement against PyTorch weights, with on-device latency well within the 150 ms mobile budget.

6.3 Multi-Crop Fact Base Generalization

Table 5 demonstrates the extension of the structured fact base across Potato, Maize, and Rice.

All fact base rows successfully passed automated cryptographic validation and dosage boundary assertions against national pesticide registries.

6.4 Dialect Robustness and Safety Invariance Audit

Across regional dialect and romanized queries (protocols E4–E6), paired normalization maintained safety non-inferiority ($\Delta_{\text{safety}} \leq 0$) with a mean Recall@10 of 0.884 across all test registers. Continuous integration replay across 50 pinned golden cases achieved a 100% invariant compliance rate (0 safety breaches, 100% unanswerable query abstention). The entire backend test suite maintains a **542 passed / 0 failed** status.

Table 5: **Verified Agronomic Fact Base Composition and Build-Time Dose Validations.**

Crop	Target Pathogen / Pest	Active Ingredient	Dose Range	Unit	PHI	Source Citation
Potato	Late Blight (<i>P. infestans</i>)	Mancozeb 80 WP	2.0 – 2.0	g/L	14 days	BARI Potato Handbook 2020
Potato	Late Blight (<i>P. infestans</i>)	Metalaxyl + Mancozeb	1.5 – 2.0	g/L	14 days	BARI Plant Pathology 2022
Maize	Fall Armyworm (<i>S. frugiperda</i>)	Spinosad 45 SC	0.4 – 0.4	ml/L	14 days	DAE & BARI FAW Guide 2021
Rice	Rice Blast (<i>M. oryzae</i>)	Tricyclazole 75 WP	0.75 – 0.75	g/L	21 days	BRRI Adhunik Dhan 2024
Rice	Brown Planthopper (<i>N. lugens</i>)	Pymetrozine 50 WDG	0.6 – 0.6	g/L	21 days	BRRI Pest Guide 2023

7 Discussion

7.1 Relational Binding vs. Substring Matching

The critical vulnerability of lexical dosage verification lies in its inability to enforce relational binding. A lexical matcher confirms the coexistence of chemical and numeric tokens in retrieved text but cannot determine whether the generated dosage was bound to the correct crop stage or formulation. By structuring claims into typed relational tuples, KrishokChat eliminates false acceptances on misbound claims.

7.2 Linguistic Equity and Normalization Constraints

Evaluating dialectal normalization under an explicit safety-preservation constraint ensures that regional farming communities receive accessible digital advisory without suffering degraded safety guarantees.

8 Limitations

We explicitly note the bounds of this work: (1) knowledge node coverage is bounded by the 2,135 curated national research documents; (2) vision classifiers perform crop/disease classification only and do not perform bounding-box localization; and (3) this study evaluates technical verification and system safety, making no longitudinal causal claims regarding field-level farmer economic returns.

9 Ethical Considerations and Governance

KrishokChat enforces privacy-by-design through strictly local-only audit logging with automated PII regex scrubbing. Agrochemical inquiries involving acute poisoning hazards or banned

substances are intercepted deterministically and routed to the national helpline (16123) and emergency healthcare contacts.

10 Reproducibility and Data Availability

All evaluation splits, benchmark datasets, and model export scripts are versioned with SHA-256 digests in the repository under `research_artifacts/` and `dataset_release/`.

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